Diagnostic performance of digital cognitive tests for the identification of MCI and dementia: A systematic review

Author links open overlay panelJoyce Y.C. Chan [a], Sarah T.Y. Yau [b], Timothy C.Y. Kwok [a c], Kelvin K.F. Tsoi [b c]

https://doi.org/10.1016/j.arr.2021.101506Get rights and content

Highlights

- Twenty-two digital cognitive tests showed a good diagnostic performance for dementia such as the SATURN.
- Eleven digital cognitive tests showed a good diagnostic performance for MCI such as the Brain Health Assessment.
- The diagnostic performances of most of the digital cognitive tests were comparable with the paper-and-pencil tests.
- All the digital tests only had a few validation studies to verify their performance, so further research is needed.

Abstract

Background

The use of digital cognitive tests is getting common nowadays. Older adults or their family members may use online tests for self-screening of dementia. However, the diagnostic performance across different digital tests is still to clarify. The objective of this study was to evaluate the diagnostic performance of digital cognitive tests for MCI and dementia in older adults.

Methods

Literature searches were systematically performed in the OVID databases. Validation studies that reported the diagnostic performance of a digital cognitive test for MCI or dementia were included. The main outcome was the diagnostic performance of the digital test for the detection of MCI or dementia.

Results

A total of 56 studies with 46 digital cognitive tests were included in this study. Most of the digital cognitive tests were shown to have comparable diagnostic performances with the paper-and-pencil tests. Twenty-two digital cognitive tests showed a good diagnostic performance for dementia, with a sensitivity and a specificity over 0.80, such as the Computerized Visuo-Spatial Memory test and Self-Administered Tasks Uncovering Risk of Neurodegeneration. Eleven digital cognitive tests showed a good diagnostic performance for MCI such as the Brain Health Assessment. However, all the digital tests only had a few validation studies to verify their performance.

Conclusions

Digital cognitive tests showed good performances for MCI and dementia. The digital test can collect digital data that is far beyond the traditional ways of cognitive tests. Future research is suggested on these new forms of cognitive data for the early detection of MCI and dementia.

Background

Dementia is a syndrome in which there is a deterioration of cognitive functions and the ability to perform daily activities (World Health Organization, 2019). Mild cognitive impairment (MCI) is recognized as the intermediate stage of dementia and normal aging (Petersen, 2011). Cognitive decline such as memory loss, deterioration of executive functions, and visuospatial abilities are often the initial symptoms of MCI and early dementia. Many cognitive tests are available for the screening of the

Methods

This study was performed according to the standard guidelines for the systematic review of diagnostic studies, including Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) (Moher et al., 2009) and guidelines from the Cochrane Diagnostic Test Accuracy Working Group (Leeflang et al., 2008, Macaskill et al., 2010).

Literature search and study selection

A total of 4581 titles and abstracts were identified, and 75 of them were extracted from the bibliographies or review articles. All titles and abstracts were screened, and 529 articles were further evaluated. A total of 473 articles were excluded (Cohen's Kappa statistics at 86% between the investigators) due to the following reasons: 15 studies were review articles; 264 studies did not evaluate a digital cognitive test; 150 studies were not validation study or lack of diagnostic results; 38

Discussion

This systematic review studied digital cognitive tests for MCI and dementia in older persons. Thirty digital tests showed a good diagnostic performance with sensitivity and specificity of over 80% for the detection of MCI or dementia. Most digital cognitive tests have a comparable or even better diagnostic performance than the traditional paper-and-pencil cognitive test.
Traditional cognitive tests usually used scoring on several questions to assess cognitive functions. Digital cognitive tests

Conclusions

Digital cognitive tests showed good diagnostic performances for MCI and dementia. Most digital cognitive tests have a comparable or even better diagnostic performance than the traditional paper-and-pencil cognitive test. The digital cognitive test is a chance to collect more digital data for improving the accuracy of classification of MCI and dementia. Future research is suggested on these new forms of cognitive data for the early detection of MCI and dementia.

Funding source

This is a self-initiated study without funding.

Conflict of Interest Disclosure

There is no conflict of interest to be declared.